

CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

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Author contribution statement

G.S. designed the dataset and the label scheme, implemented the processing pipeline and the quality-control procedures, and drafted the manuscript. A.S., A.T., M.K., R.C., D.H., F.M., H.S., M.S., S.S., M.M. and J.K. performed annotation correction and quality-control review of the released segmentations under supervision. M.I. and N.A. performed the radiological reads — the Castellvi grading and the adjudication of candidate instrumentation. [BEFORE SUBMISSION: confirm each contribution individually with the author concerned and expand to CRediT roles.] All authors read and approved the final version of the manuscript.

Conflict of interest statement

The authors have no relevant conflicts of interest to disclose.

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Data and code availability (unblinded)

Dataset, archived before submission: <https://doi.org/10.5281/zenodo.22139642> (concept identifier, resolves to the current version); the version described in the manuscript is v7, <https://doi.org/10.5281/zenodo.22242745>.

Code: <https://github.com/Gregory-Schwing-MD-PhD/CTSpinoPelvic1K>, tagged at the release commit.

Mirror: <https://huggingface.co/datasets/OpenSpineConsortium/CTSpinoPelvic1K> — a convenience mirror, not the archive of record.

Ethics

This work uses publicly available, de-identified imaging and annotations derived from it. The Wayne State University Institutional Review Board issued a Notice of IRB Administrative Determination of Non-Human Participant Research on 2 September 2026 (WSU IRB HPR Number 2026-269, “Open Spine Consortium: Secondary Analysis of Publicly Available, De-Identified Imaging Datasets and the Derived Annotations, Software, and Models Produced From Them”), finding that the project does not constitute human participant research under the Common Rule at 45 CFR 46 and that IRB review and oversight are not required.

**CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated
for lumbosacral transitional anatomy**

[Author names removed for double-anonymized review]

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Purpose: Lumbar levels cannot be numbered reliably when a lumbosacral transitional vertebra (LSTV) is present. Levels are conventionally established by counting downward from C2, the reference standard for numeration¹. An abdominal field of view lacks that landmark, leaving a specific ambiguity: four rib-free lumbar vertebrae may mean an L1 bearing a lumbar rib or an L5 assimilated to the sacrum, and six may mean a true sixth lumbar vertebra or a T12 whose ribs are aplastic². CT-SpinoPelvic1K is built to test whether local morphology can resolve them without counting from the C2 anchor. It provides 802 CT records anchoring each lumbar level on the lowest rib-bearing vertebra above and the first sacral segment below, with radiologist-sourced spine *and* pelvic annotation on one frame, per-level ribs and femora. Classes for a sixth lumbar vertebra, a thirteenth thoracic vertebra, a separate first sacral segment, and lumbar ribs make the set expressive enough for anatomical anomalies.

Acquisition and Validation Methods: Records pair TCIA CT COLONOGRAPHY imaging with CTSpine1K³ vertebral and CTPelvic1K⁴ pelvic labels on the acquisition with highest bone coverage, under a VerSe-native scheme. Validation: geometric invariants (802/802 pass); rib-vertebra incidence across 5,749 evaluable ribs (2 offsets, 0.035%); and agreement of spinopelvic measures with published values.

Data Format and Usage Notes: Gzipped NIfTI image/label pairs sharing one affine, canonicalized to PIR, with frozen patient-grouped LSTV-stratified five-fold splits and a reference loader; archived at 10.5281/zenodo.22139642.

Potential Applications: Testing whether a network can classify a vertebra from local features alone; updating textbook morphometry drawn from small cadaveric series; spinopelvic assessment at the lumbosacral interface; variant studies; opportunistic screening. *Limitations:* thoracic ground truth is field-of-view limited; postural angles supine; no held-out test set; the rib layer is a triaged-review pseudolabel; Castellvi grades assigned by two radiology resident physicians.

6 I. INTRODUCTION

7 The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, and
8 a vertebra is identified by where it falls in that sequence. Transitional variants sit at the two
9 ends of the lumbar column — the thoracolumbar transitional vertebra (TLTV) above, the
10 lumbosacral transitional vertebra (LSTV) below — and they disturb identification in two
11 ways at once. They change what the border vertebra and its ribs *look like*: a thoracolumbar
12 border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral
13 one may be partly or completely assimilated to the sacrum, with an anteriorly wedged body,
14 a reduced disc beneath it and hypoplastic facet joints, or conversely separated from it with
15 a squared body, lumbar-type facets and a full-height disc². They also change *how many*
16 vertebrae each region contains — eleven or thirteen thoracic, four or six lumbar.

17 Either change alone would be tractable. Together they make it difficult to state defini-
18 tively what any vertebra at these borders *is*. Four rib-free vertebrae above the sacrum may
19 mean an L1 bearing a lumbar rib or an L5 assimilated to it; six may mean a true sixth
20 lumbar vertebra or a T12 whose ribs are aplastic. Each pair yields the same count, and the
21 morphology that would separate the two readings is not decisive to a human reader in every
22 case. Whether it nonetheless differs enough for a learned classifier to separate them, or at
23 least to assign each reading a probability, is the question this release is built to support.
24 Absent that, what remains is the sequence — conventionally established by counting down-
25 ward from C2, the reference standard for numeration¹, and unavailable in any spine-limited
26 acquisition. LSTV is common, reported between 4% and 30% depending on definition, and
27 wrong-level surgery is its most serious consequence.

28 Existing public collections do not make this tractable, and not because they are small
29 (Fig. 1): the spine collections omit the pelvis, the pelvic ones carry no vertebral count, and
30 the whole-body schemes have neither an L6 class nor a class for a rib borne by a lumbar
31 vertebra, so a six-lumbar spine cannot be written down in them at any segmentation quality.
32 VerSe^{5,6} is the instructive case, and Fig. 1(b) gives it: a collection can enrich for anatomical
33 variants deliberately, grade them, and still be unable to represent the ones it grades.

34 CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and
35 gives a class to each structure an enumeration anomaly produces: a sixth lumbar vertebra,
36 and a rib borne by a lumbar vertebra. A scheme without those classes must record such

37 anatomy as something it is not.

38 *Where the material is.* Everything described here is v7 of a single archived dataset,
39 deposited at <https://doi.org/10.5281/zenodo.22242745>. Citations should use <https://doi.org/10.5281/zenodo.22139642>, the permanent identifier for the dataset across all
40 of its versions, which resolves to whichever version is current; the version identifier above
41 pins the exact release measured in this manuscript. Documentation, the label scheme, the
42 reference loader and the per-record quality-control tables are distributed with the archive
43 and are described in Sec. III.

45 I.A. Vertebra labeling when the scan cannot settle the count

46 *The limitation, stated plainly.* No scan in this corpus contains C2 or the whole cervical
47 spine, so the conventional count cannot be performed. That is a property of abdominal
48 imaging rather than of the annotation, and it means a variant at the thoracolumbar junction
49 cannot be *resolved by counting* here at all: a thirteenth thoracic vertebra, a lumbar rib, and
50 a *stump rib* — a hypoplastic twelfth rib, one of the cardinal indicators of a thoracolumbar
51 transitional vertebra — produce overlapping appearances.

52 *What the release does instead.* No vertebral label is asserted where it cannot be supported.
53 Every quantity is expressed against the two landmarks present in essentially every abdominal
54 scan — the sacrum below and the lowest rib-bearing vertebra above — so the description is
55 positional rather than nominal: a vertebra by how many rib-free vertebrae separate it from
56 the sacrum, a lowest rib by its length as a fraction of the rib above it, a transverse process
57 by its span and its distance from the ala. None of these changes according to whether a
58 reader would have said L5 or S1, so cases that disagree about the name remain comparable.
59 Where the junction is in view the count is determined and the conventional name follows,
60 and the two descriptions agree.

61 *And the morphology may be local rather than global.* Counting is not the only route to
62 an identity. A thoracic vertebra differs in shape from a lumbar one — in the span of its
63 transverse processes relative to its body, in the proportions of its endplates and canal — and
64 if that difference holds at the boundary itself, the phenotype is decidable from the vertebra
65 in front of the reader without any global count. On this corpus it does hold: separating
66 T12 from L1 on shape alone, with each patient’s own size divided out and cross-validation

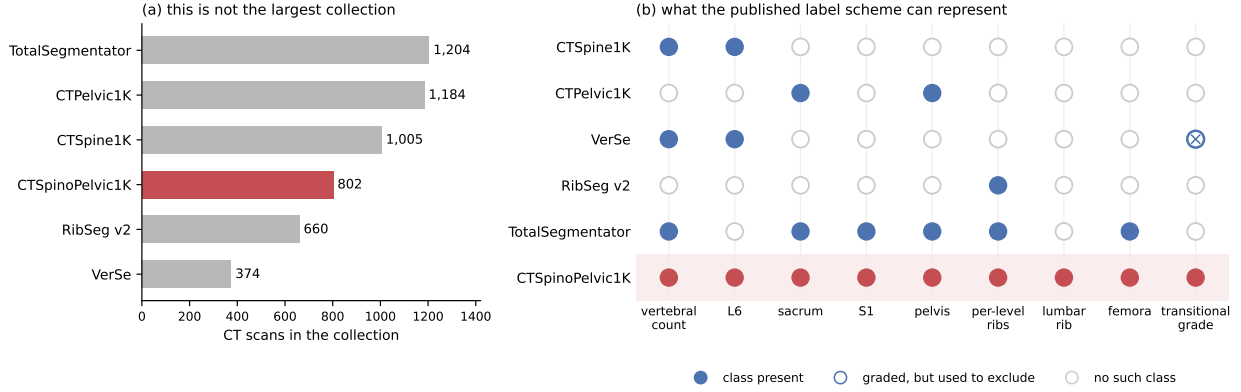


FIG. 1 Public CT collections at the lumbosacral junction. (a) Size, stated first because it is the claim this dataset does *not* make: three of the five comparators carry more scans, and RibSeg v2⁷ labels more ribs. (b) What each published scheme can *represent*, a property of its class list rather than of its segmentation quality. VerSe is graded-but-excluding: it applies the Castellvi classification and uses it to decide what to omit, leaving vertebrae partially fused to the sacrum unsegmented⁶ — 25 of the 33 graded records here.

grouped by case, gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. Size is deliberately removed, because a classifier given raw millimeters separates thoracic from lumbar immediately and has learned nothing that helps at the junction, where the two are nearly the same size.

That is reported as a property of the released measurements rather than as a method: the information needed to settle these variants is present locally.

II. ACQUISITION AND VALIDATION METHODS

II.A. Sources, and why a crosswalk was necessary

Both CTSpine1K³ and CTPelvic1K⁴ draw part of their imaging from the TCIA CT colonography collection^{8,9}, and the vertebral and pelvic annotations in both were produced under board-certified radiologist supervision. Neither released the mapping from an annotation to the specific CT series it was drawn on.

That mapping is not a bookkeeping detail. Every patient in CT COLONOGRAPHY was scanned *twice*, prone and supine, often with more than one kernel, so one patient identifier

81 resolves to several volumes differing in posture, position, field of view and noise — and an
82 annotation carries a patient identifier and nothing finer.

83 Placing it on the wrong volume of the same patient is not merely a displacement. The
84 spine is mildly deformable, and turning a patient from supine to prone alters lumbar lordosis
85 and the alignment of each segment against the next, so a vertebra moves by a different
86 amount, and in a different direction, from the one above it. Outlines drawn on one series
87 are the wrong *shape* for the other, and no rigid realignment recovers them. In isolation such
88 a mask looks like competent work; it reveals itself only when overlaid on the image.

89 The crosswalk (Fig. S1) therefore resolves each annotation first to a patient and then,
90 within that patient’s volumes only, to the series whose bone *agrees* with it: candidates are
91 thresholded at bone attenuation and scored by overlap with the annotation mask. Bone
92 rather than image similarity, because bone is what the annotation describes — two acqui-
93 sitions of one abdomen resemble each other everywhere except in where the skeleton sits.

94 II.B. Fused and split records

95 The two annotation sets were placed independently, so they do not always land on the
96 same series. Of 802 records, 342 carry spine and pelvic annotations on the same volume
97 (*fused*), 351 are *separate*, 89 are *spine_only* with no pelvic annotation for that patient, and
98 20 are *pelvic_only*. These four are the values of the manifest’s `match_type` field; the coarser
99 `config` field folds *separate* into *spine_only*, so a user wanting the four types should read
100 `match_type`. The split is the crosswalk’s main finding rather than bookkeeping: in those
101 separate records — nearly half of all patients — the two public collections annotated *different*
102 *volumes of the same person*, and neither recorded which. A pelvic annotation placed on the
103 other acquisition of a prone/supine pair cannot be combined with the spine annotation
104 without resampling across a change of posture. Those records are retained rather than
105 discarded, and held out of the postural analyses, because two acquisitions of one patient
106 differing in spinal alignment are a resource for a mobility study rather than a defect.

II.C. Densification, and why it is asymmetric

A corpus in which half the records lack a pelvis is not usable as a spinopelvic dataset, so the missing structures were pseudolabeled — asymmetrically (Fig. S1), because the two directions are not equivalent.

Pelvis onto spine-only records is the safe direction. The sacrum and innominate bones are single objects with no enumeration: no count to get wrong, no naming convention to choose, no analog of a transitional vertebra. Quality is reported as held-out performance on the *pelvic-only* records, withheld from training for exactly this purpose.

The reverse direction carries the whole problem this dataset exists to address, since a pseudolabeled spine must commit to a vertebral count and on a transitional vertebra will commit to the commoner reading. Those 20 records were therefore corrected by hand rather than densified automatically — tractable at twenty cases, which is the practical reason the asymmetry runs the way it does.

II.D. Ribs

No public rib annotation covers this cohort, and the two available automatic sources fail in complementary ways.

Möller’s binary rib network¹⁰ is a detection model of high reported accuracy, but it segments ribs that lie *entirely within* the field of view; a rib clipped by the edge of an abdominal acquisition is liable to be missed altogether. Since this is a colonography cohort, the lower ribs are routinely clipped, and those are precisely the ribs the count depends on.

TotalSegmentator¹¹ does segment clipped ribs and supplies the per-rib numbering that a binary network cannot. Its characteristic failure is at the other end of the bone: the most medial portion of the rib — roughly the last tenth to sixth approaching the costovertebral joint — is frequently absent. A rib truncated at its medial end never reaches the vertebra it articulates with, so the incidence test that assigns a rib to its level has nothing to test.

The two were therefore unioned (Fig. S1): TotalSegmentator supplies numbering and the medial-clipped cases, Möller supplies detection where its output is more complete, and the union carries TotalSegmentator’s per-level identities.

The result is a pseudolabel whose review was triaged rather than exhaustive, which is

worth stating plainly because a layer described only as “human-corrected” invites the reader to assume every bone was looked at. A label-based rule flagged every record in which one rib bone carried more than one identifier, or in which a rib failed to reach the vertebra it articulates with — a single bone with two numbers being a numbering error by construction, whatever the anatomy. The rule selected 152 records; 148 were reviewed and corrected, and 4 were not and are identified in the release. Reviewers were medical students working through OpenSpineConsortium¹², trained against a written labeling protocol, with every correction recorded against the annotator who made it. Records passing the rule were not individually inspected, so the layer’s guarantee is that it is free of the failure modes the rule detects, and no stronger. The residual rib–vertebra offsets reported in Sec. II.G are the independent check on what the rule missed.

II.E. Label scheme and counting anchors

The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26), and every non-VerSe structure takes a fixed identifier above that range — S1 carved from the sacrum (29), hips (30–31), femora (32–33), ribs (34–45 left, 46–57 right), soft tissue (58–73), lumbar ribs (74–75) and surgical hardware (76–82).

Two anchors make the count decidable, and Fig. 2 shows both: the lowest rib-bearing vertebra above, **S1** carved from the sacrum below. Each earns its place by what it supports rather than by being an endpoint — S1 supplies the sacral endplate landmark that sacral slope and pelvic incidence are measured from, and with both brackets present L5 and L6 are distinguishable where a spine-limited view alone would not be.

How S1 is cut. The sacrum’s outer boundary is the source annotation and is never altered; S1 is the cranial part of that same bone, separated by a plane. The plane’s normal is the sacrum’s own cranio-caudal axis, made orthogonal to a left–right axis that is measured rather than assumed — the normal of the reflection plane best mapping the sacrum onto itself, which removes the patient’s rotation within the scanner. That rotation is not negligible: median 4.5°, maximum 16.1°, and 508 of 801 records above 3°. The level of the cut preserves the sub-division volume of the preceding release, so the plane changes the shape and orientation of the S1/S2 boundary and not how much of the sacrum is called S1.

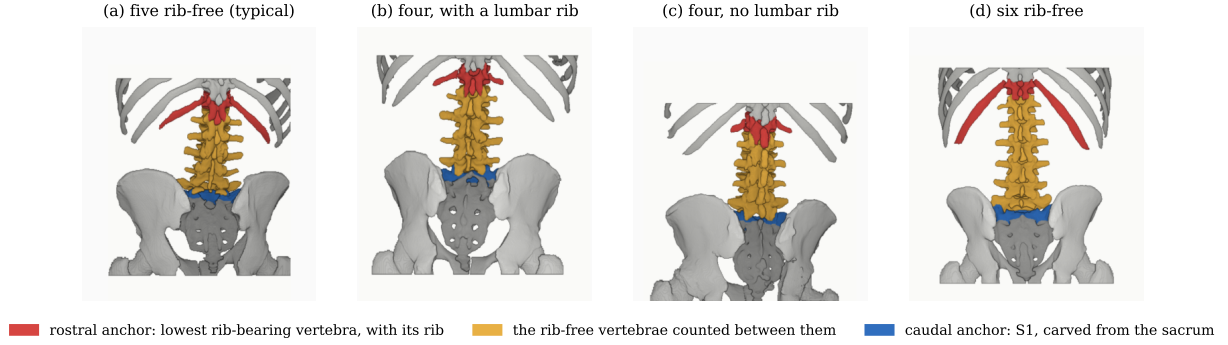


FIG. 2 The two anchors and the interval between them, from the released labels, posterior view. Rostral anchor: the lowest rib-bearing vertebra and the rib that makes it one. Caudal anchor: S1. Both are found from the labels rather than assumed, and panels are colored by role rather than identifier, since coloring T12 would assert the count the figure derives. Panels (b) and (c) reach the same interval of four from opposite ends and are indistinguishable by it. All panels are at one millimeter scale, aligned at the sacral base.

166 Per-record geometry ships with the archive.

167 Neither anchor is novel: TotalSegmentator¹¹ carries a separate S1 class and per-level
 168 ribs, and VerSe⁵ carries L6. Because this release uses the VerSe identifiers, its L6 *is* VerSe's,
 169 interoperable rather than local to this work. L6 is retained not for novelty but because a
 170 scheme without it cannot record a six-lumbar spine at all, and must renumber the column
 171 or drop a level to fit.

172 **A rib on a lumbar body receives its own class**, and this is the one class here with no
 173 published counterpart. A lumbar rib and a stump rib are the same object under two counts,
 174 so assigning it a number would commit the label to one reading of an enumeration anomaly.
 175 A scheme that numbers every rib 1–12 has nowhere to put a thirteenth: the annotator must
 176 either call it rib 12 — asserting the vertebra beneath it is thoracic, the very question at
 177 issue — or discard it. TotalSegmentator carries ribs 1–12 per side and no such class, and
 178 VerSe's T13 is a vertebra rather than a rib. 16 released records carry a lumbar rib and 18
 179 carry an L6.

180 Figure 2 shows why the interval is the primitive and the label is derived. Among the
 181 records with four rib-free vertebrae, some reach that count through a lumbar rib and some
 182 without one; the two are indistinguishable by the count alone and are distinguished here

183 only because rib presence is itself labeled.

184 Sixteen records carry a lumbar rib in the released volumes: thirteen bilateral and three
185 unilateral, all three on the right. That laterality is reported because it is what the release
186 contains and not as a finding — sixteen records cannot support a statement about side
187 preference, and none is made.

188 Eighteen records carry an L6. Fourteen are labeled lumbarization, two sacralization,
189 one semi-sacralization and one normal, which is worth stating because it shows the two
190 annotation axes disagreeing in the open: a record can carry six lumbar-type vertebrae and
191 still be read as a sacralization, or as unremarkable, depending on where the reader started
192 the count.

193 Both counts are derived from the released label volumes, and so are the manifest fields
194 that report them. That is a correction rather than a design note: through v5 the `has_l6`
195 flag was true for exactly one record, which contains no L6, and false for all eighteen that
196 do, while `has_lumbar_rib` was false everywhere and `n_lumbar_labels` read zero in 799 of
197 802 records. A reader selecting the six-lumbar cases on those fields would have received one
198 record that is not one, and a reader selecting lumbar ribs an empty set, with nothing to
199 indicate either. The fields are rebuilt for this release by counting identifiers 25 and 74–75 in
200 each volume, and `lumbar_rib_side` is added alongside them.

201 II.F. Computational tools

202 *Access.* Every processing step described above is performed by code released under an
203 open-source licence and archived with the dataset; no step depends on software that is
204 not publicly obtainable. The release archive carries the reference loader, the label-scheme
205 definition and the quality-control scripts that generated the tables in this manuscript.

206 *Versions and parameters.* Segmentation of femora, the sacral sub-division and the per-
207 level rib numbering used TotalSegmentator¹¹ (`total` task, release 2.x) on a single graphics
208 processing unit. The binary rib network is Möller’s¹⁰ released weights, applied through the
209 nnU-Net v2 framework¹³. The pelvic completion for records whose pelvis was absent used a
210 five-fold nnU-Net v2 ensemble, applied out-of-fold: each record is completed by the fold that
211 never trained on it, so no record is completed by a model that has seen it. Image handling
212 throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is

TABLE I Release validation. Gates run in order and a failure stops the chain.

Check	Result
Geometric invariants (all cases)	802/802 pass
label shape, affine, spacing match CT	—
no identifier outside the scheme	—
left/right structures correctly sided	—
Ribs present	11,548
not evaluable (expected vertebra outside the FOV)	5,788
no vertebra within the anchor distance	11
evaluable	5,749
matching the expected vertebra	5,747
residual offset	2 (0.035% of evaluable)
misnumbered cases	0
Spinopelvic medians within published range	6/6

213 written, so every label shares its image’s grid and affine exactly.

214 *Generative artificial intelligence.* Large-language-model assistance was used for manuscript
215 preparation — drafting and editing of text, and generation of analysis and figure code —
216 under author review. It was not used to generate, annotate or interpret any imaging data,
217 and no scientific claim in this manuscript originates from it. All numbers reported here are
218 computed by the released code from the released volumes.

219 II.G. Validation

220 Validation is layered, and the ordering matters: measurements computed on a corpus
221 that has not established internal consistency do not look broken, they look finished.

222 *Geometric invariants.* Every case is checked for label geometry agreeing with its CT in
223 shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty
224 label; and for sided structures falling on the correct sides, with the left–right axis read from
225 the affine rather than assumed. The check exists because a renumbering pass once wrote
226 a reoriented array under a canonical affine and transposed a label away from its image, a

227 failure invisible downstream.

228 The check compares *each sided pair separately*, and both simpler alternatives miss cases
229 it catches. Testing the ribs alone passed all 802 records while four of them carried `left_hip`
230 on the patient’s right. Pooling all sided structures into one test recovered three of those four
231 and missed the fourth, because hips and femora outweigh the ribs by voxel count, so a large
232 correctly-sided structure masks a smaller transposed one. A gate that passes everything is
233 not evidence of a clean corpus until it has been shown capable of failing.

234 *Rib-vertebra incidence.* Each rib is matched to its nearest vertebra and compared against
235 the vertebra its number implies; the chain is given in Table I. The denominator is stated
236 there rather than only the numerator because quoting the two residual offsets against all
237 11,548 ribs would halve the rate by counting ribs the check never examined — and because
238 more than half the ribs in a screening abdominal CT being unevaluable is itself a fact
239 about the corpus worth reporting. Both offsets are field-of-view truncations on individually
240 characterized cases, one at +1 level and one at −1, and are reported rather than suppressed:
241 a threshold tuned until the count reaches zero yields a cleaner number and a less informative
242 one.

243 The same check carries an invariant that reads as a null result. Of the 5,749 evaluable
244 ribs, *zero* are assigned to a lumbar vertebra — not because no lumbar ribs exist, since 16
245 records carry one, but because such a rib takes class 74 or 75 and never enters the numbered
246 set the test examines. A numbered rib landing on a lumbar vertebra would be a counting
247 error of exactly the kind this dataset exists to expose.

248 *Agreement with published values.* Derived spinopelvic measures are compared against
249 Vialle *et al.*¹⁴ (Table II). Pelvic incidence is the strongest of these checks because it is a
250 morphological property of the pelvis rather than a posture, and so is directly comparable to
251 a standing reference cohort.

252 II.H. Castellvi typing

253 Every record whose vertebral count departs from five carries a Castellvi grade¹⁵ in the
254 released manifest: 33 cases, typed I–IV with the *a/b* unilateral–bilateral qualifier. This is
255 released as an annotation layer because the grade and the count are *different axes* and the
256 dataset is built to show that they are: grade IIIB occurs here at rib-free counts of four, five

TABLE II Derived measures against a published asymptomatic reference ($n = 260$, standing)¹⁴. This cohort is supine, and the two postural measures depart in opposite directions — sacral slope 4.1° lower, pelvic tilt 4.3° higher. That is one discrepancy, not two: $PI = PT + SS$ holds by construction, so with pelvic incidence at its published value any fall in one appears as an equal rise in the other.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	54.7 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis (supine)	52.7°	43 ± 11.2 (standing)
PI – LL mismatch	2.6°	centered near 0

and six alike, and seven of the 33 graded cases carry a perfectly normal count of five. A corpus that recorded only the count would describe those seven as unremarkable.

The grades were assigned by two radiology resident physicians. The distinction the classification turns on — bony fusion against articulation without it² — is the one carrying clinical weight.

II.I. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here for one reason: **an iatrogenic fusion is indistinguishable from a congenital one to a distance measurement**. A cage-bridged interspace reads as “no gap” exactly as a congenitally fused transitional vertebra does, so an unrecorded instrumented case is a false positive on the central measure — and identifiers 76–79 were declared in every prior version and populated in none.

II.I.0.a. Detection over-calls by a factor of seven. A threshold at 1800 HU inside a shell around the labeled skeleton flags 84 of the 802 records. Raising it to 2500 HU — the lower of the two values validated in the metal-segmentation literature — leaves 52 with a component above a 40-voxel floor. One of the two radiology resident physicians read all 52 and confirmed 11, rejecting 41 (79%) as contrast, calcification and reconstruction artefact. Instrumentation

prevalence in this cohort is 1.4%, not the 10.5% an unreviewed threshold reports.

II.I.0.b. Saturation does not separate implant from artefact. Both groups reach the scanner ceiling and 4 rejected proposals *exceed* it, peaking at 11,798 HU — the signature of reconstruction overshoot rather than of denser metal. What separates them is position and size together: a real surgical site *and* volume, every confirmed implant at 2,586 mm³ or above and every rejection at 1,768 mm³ or below.

II.I.0.c. The subtype block had to be extended. Identifiers 76–79 name spinal instrumentation — generic, cage, screw-and-rod, plate — and cannot describe what this cohort holds, so **80 arthroplasty**, **81 sacroiliac screw** and **82 osteosynthesis** were added. The distinction is clinical, not cosmetic: osteosynthesis holds parts of the same bone together where arthroplasty replaces a joint, and without it the shape rules call a femoral stem a rod, since a stem is long and thin and that is all “linear” tests. The confirmed cases comprise 8 arthroplasties, one osteosynthesis, one sacroiliac screw fixation and one pair of threaded cylindrical interbody cages.

II.I.0.d. Metal outranks bone, and that is a subtraction. In every confirmed case the implant was already inside a bone label, a segmenter handed a bright object against a cortical surface having taken it for bone. Naming the hardware therefore reclaims voxels rather than adding them: 1,538,852 voxels across the eleven records. The consequence reaches past bookkeeping. Pelvic incidence and pelvic tilt are measured from the femoral head center, and in 8 of these cases that head is an implant — any spinopelvic parameter previously computed from them was measured on metal.

III. DATA FORMAT AND USAGE NOTES

Each record is a gzipped NIfTI image/label pair sharing an identical 4×4 affine, canonicalized to PIR, requiring no resampling. Masks are NIfTI rather than DICOM: the policy prefers DICOM where applicable, and for volumetric segmentation masks NIfTI is what the downstream tooling expects.

Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the data as `splits_5fold.json`. The five validation folds partition all 802 records, so the release provides cross-validation and *not* a held-out test set. Users reporting a single held-out number should carve one out and state which records it holds, because the folds shipped

TABLE III Key data types and metadata, and the fraction of the 802 released records for which each is populated. Counts are obtained by reading the released manifest, not by assertion. A field below 100% is not an error: age and sex are absent where the source collection did not publish them, a pelvic series identifier exists only where a pelvic annotation was placed, and a Castellvi grade exists only for the transitional records that were graded.

Field	Records	Available
Identifier, image path, label path, configuration, match type	802	100.0%
Patient position, tube potential, section thickness, kernel	802	100.0%
Transitional label and class	802	100.0%
Spine and pelvic annotation origin	802	100.0%
Instrumentation and partial-annotation flags	802	100.0%
Image/label alignment check	802	100.0%
Spine series identifier; bone-fraction check	782	97.5%
Scanner manufacturer and model	768	95.8%
Sex	749	93.4%
Age	709	88.4%
Pelvic series identifier	713	88.9%
Castellvi grade	33	4.1%

here will not do that job.

Stratification is on the transitional subtype rather than on a binary LSTV flag, and it enforces a minimum of each rare subtype in every validation fold: each fold’s validation set carries three sacralization and three lumbarization records, so no fold is evaluated on a cohort missing the class the dataset exists for. With 15 lumbarization and 15 sacralization records across 802, that is the practical limit of what stratification can guarantee, and it is why a fold-level metric on the rare classes carries an error bar far wider than its own decimal places.

Table III lists the key data types and metadata against the fraction of records for which each is populated. Two entries are worth reading carefully. The Castellvi grade is present for 4.1% of records because it is defined only for the transitional ones; among those it is complete. A pelvic series identifier is present for 88.9% because the crosswalk records it

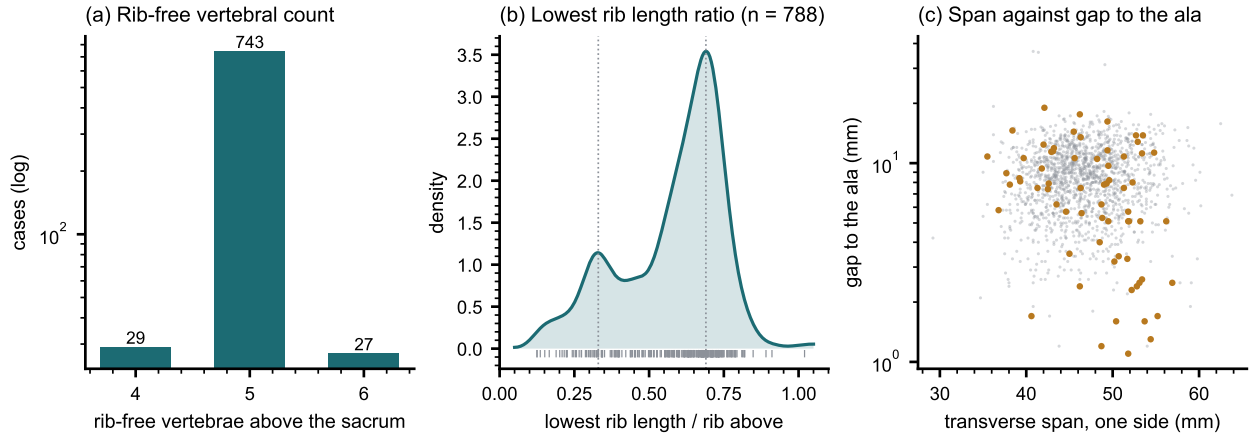


FIG. 3 Count-free measures, valid without knowing which vertebra is which. (a) Vertebrae between the lowest rib and the sacrum: an interval, not a level assignment. (b) Lowest rib length as a fraction of the rib above, showing two populations near 0.33 and 0.69 rather than one distribution with a tail. (c) Lowest lumbar transverse span against its gap to the sacral ala; cases carrying a source transitional label are marked.

only where a pelvic annotation was actually placed, and users joining on it should expect the shortfall rather than treat it as loss.

The archive of record is Zenodo (10.5281/zenodo.22139642). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM’s *other* value, a recorded value and not a missing one, and 53 carry no sex field. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals, and records outside the female/male dichotomy are excluded from sex-stratified measures — a limitation of those measures rather than of the cohort, stated because folding eleven people into “unrecorded” would misdescribe the source. The lower age bound is the screening guideline rather than a selection applied here, and it is why the measures below sit closer to a reference range than a surgical series would.

Count-free measures (Fig. 3). Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which of the two is present does not by

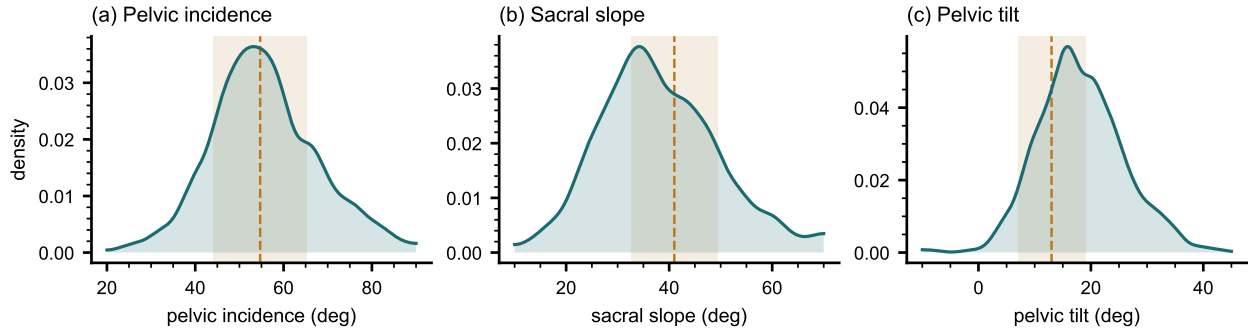


FIG. 4 Spinopelvic measures against published reference bands ($\text{mean} \pm \text{SD}$) from an asymptomatic standing cohort.¹⁴ Sacral slope is expected to sit below a standing reference because it is postural and this cohort is supine; pelvic incidence is not postural, which is why it can be compared without apology and why its agreement is the strongest check here.

332 itself determine what to call it. The lowest-rib length ratio is bimodal.

333 *Level gradients and spinopelvic measures* (Fig. 4). Vertebral dimensions increase caudally
 334 in step with the load each level carries. Pelvic incidence does not vary with age, while pelvic
 335 tilt increases — the descriptive signature of a pelvis retroverting as a spine ages.

336 *Opportunistic measures.* Every scan carries a vertebral bone density measurement at no
 337 additional dose.¹⁶ L1 trabecular attenuation is reported at the published standard site.

338 V. POTENTIAL APPLICATIONS AND LIMITATIONS

339 The principal use is the *spinopelvic interface*: spine and pelvis carry radiologist-sourced
 340 annotation on one coordinate frame in 802 records, which makes sacral morphology measur-
 341 able against the lumbar column rather than in isolation — the sacral endplate and its slope,
 342 the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geom-
 343 etry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum
 344 and femoral heads rather than from landmarks placed on a radiograph, so it is reproducible
 345 from the labels themselves.

346 The transitional layer serves that use rather than competing with it: no trajectory can be
 347 planned across a junction whose segments cannot be named. Level-numbering benchmarks,
 348 variant studies and opportunistic screening follow from the same annotation.

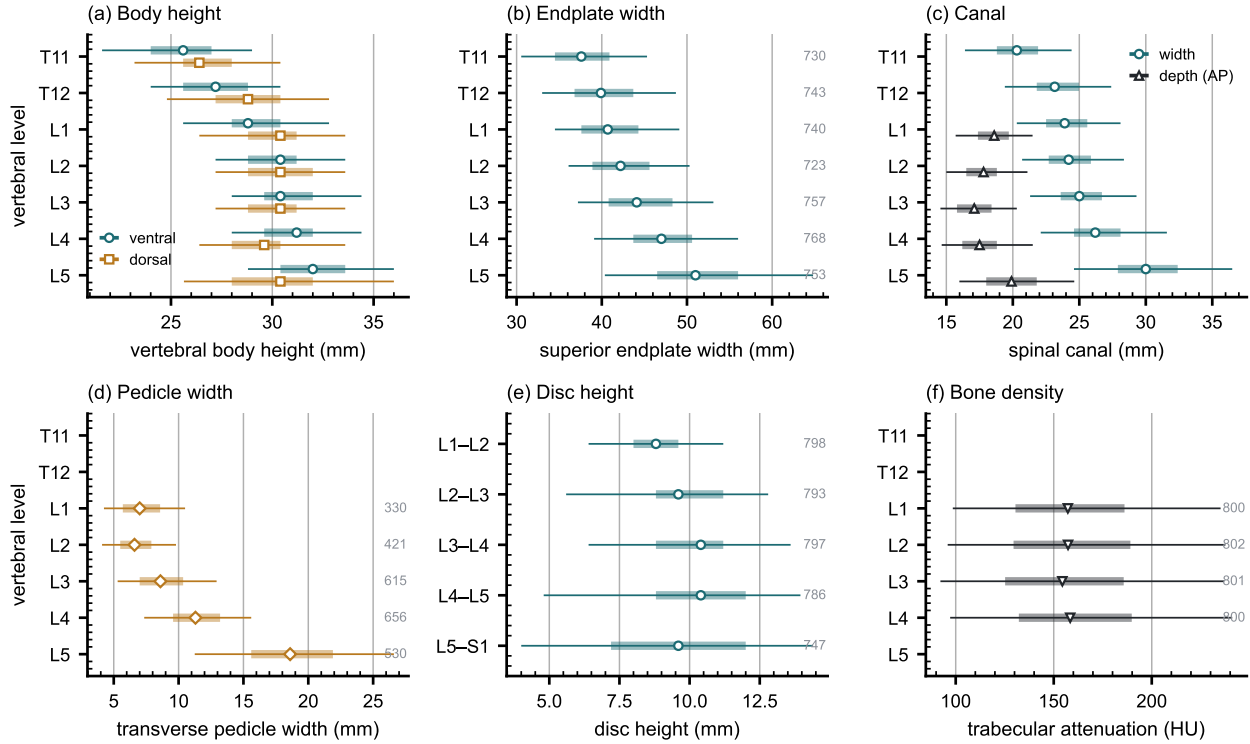


FIG. 5 Lumbar and lower thoracic morphometry by level, in the format the standard reference figures use — level on the vertical axis, measurement on the horizontal. Marker, median; bar, interquartile range; line, 5th–95th percentile; the number at the right of a row is its n . (a) Vertebral body height, ventral against dorsal. (b) Superior endplate width. (c) Canal width against depth. (d) Transverse pedicle width. (e) Disc height, drawn at the interspace it occupies. (f) Trabecular attenuation at the published opportunistic-screening site. Coverage is not uniform and is drawn rather than assumed: canal depth and pedicle width rest on 330–656 records because they need an axial extent many field-limited series do not carry, while the remaining measures rest on 720–784. No comparison between levels is tested and none is implied; the intervals are dispersion.

V.A. Three uses the released measurements support

Naming a level from its own geometry. Wrong-level spine surgery runs at roughly one in 3,110 spinal procedures, and the cause most often cited is the transitional anatomy this cohort was assembled around.¹⁷ The failure is structural rather than careless: the count is ambiguous exactly where the variant sits. Per-level body height, canal width, endplate width, transverse-process span and wedge ratio are released for T11–L5 across all 802 records, and because the labels were assigned against the twelfth rib rather than by enumeration, a

model trained on them is not learning the convention that fails.

Reference morphometry with its spread. The values a surgeon consults for vertebral body height, canal dimensions and pedicle width come from cadaveric series of a few dozen specimens, plotted as single values with no indication of spread.¹⁸ They cannot tell a reader whether the patient in front of them is unusual, because they never show what usual looks like. The same measures are given here from 802 records with the distribution attached (Fig. 5), reproducing two properties of the classical figures — body height crosses over, dorsal exceeding ventral at T11–T12 and reversing by L4–L5, and pedicle width widens steeply into the lower lumbar spine — and adding the width of each distribution, which is the part needed to judge an individual. A larger n sharpens a distribution without changing the population it came from, and this one is a screening cohort aged 50 and over.

Patient-specific models. Planning is moving toward patient-specific biomechanical models rather than population norms.¹⁹ Such a model needs spine, pelvis and femoral heads in one frame, which is this release’s construction; in 351 patients the annotation sits on two acquisitions in different positions, giving a within-patient change in alignment against which a predicted postural response can be checked.

Its limitations are part of it, and they are stated at the level of detail a reader would need to catch them independently.

Coverage. Thoracic ground truth is field-of-view limited and does not extend to T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat. The cohort is a colorectal screening population aged 50 and over, so its distributions should not be read as representative of a surgical one.

Declared but empty, and confirmed so. The soft-tissue identifiers (58–73) and the partial-annotation sentinel (255) are declared in the scheme and occur in no record — established by counting identifiers across all 802 released volumes rather than asserted. Both are retained so a later release can use them without renumbering, and their absence should be read as absence of annotation rather than absence of the structure. The hardware identifiers were in this position until v6 and are no longer: 76–82 are populated in the eleven instrumented records described in Sec. II.I.

Strength of the labels varies by structure, and the release does not average over that. Vertebral labels derive from radiologist-supervised source annotations, corrected where those sources were wrong. Pelvic labels on records that lacked one are pseudolabeled. The rib

layer is a pseudolabel whose human review was triaged by an automated rule rather than exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional labels derive from two independent sources and are not uniformly adjudicated — which is precisely why the measures reported here are count-free — and the Castellvi grades were assigned by two radiology resident physicians. The sacral sub-division inherits its level from an automatic method, and in 100 records that level places S1 above half the sacrum or below fifteen percent of it, which no first sacral segment is; those records are flagged in the released quality-control table and should be filtered before any measurement depending on the sacral endplate.

Structures are not universally present. The same census shows the release is not uniformly complete. Sacrum, hips and femora are present in all 802 records. One record carries no S1, one lacks T12, and nine lack an L5 identifier — in every case because the structure lies outside the field of view or, for L5, because the lowest lumbar segment is labeled L6 or incorporated into the sacrum. Users filtering on the presence of the caudal anchor should filter explicitly rather than assume it.

VI. DISCUSSION

Comparison with related material. Against the collections in Fig. 1, the contribution is the crosswalk placing spine and pelvis on the same series, together with classes for the anatomy an enumeration anomaly actually produces: where another scheme must record a sixth lumbar vertebra or a lumbar-borne rib as something else, this release records it as itself.

VII. CONCLUSION

CTSpinoPelvic1K places spine, pelvis, per-level ribs and femora on one coordinate frame in 802 records, and gives the anatomy that makes lumbar numbering ambiguous a class of its own rather than recording it as something it is not. Because both counting anchors are explicit, a level can be identified from a field of view that cannot support the conventional count downward from C2 — which is every record here. The release is archived under a persistent identifier, ships what is needed to verify it, and states per structure how strong

each label is.

DATA AVAILABILITY

The release described here is v7, permanently archived at <https://doi.org/10.5281/zenodo.22242745>. The dataset as a whole, across versions, is cited as <https://doi.org/10.5281/zenodo.22139642>, which always resolves to the current version. The archive carries the reference loader, the label-scheme definition and the quality-control tables, and the analysis code is distributed with it under an open-source licence. The repository URL is omitted here because it identifies the authors; it is given on the title page and will be restored to this section for the camera-ready version.

CONFLICT OF INTEREST

The authors have no relevant conflicts of interest to disclose.

ETHICS

This work uses publicly available, de-identified imaging and annotations derived from it. The authors' institutional review board determined that it does not constitute human participant research under the Common Rule at 45 CFR 46, and that review and oversight are therefore not required.

REFERENCES

- ¹J. Lian, N. Levine, and W. Cho, A review of lumbosacral transitional vertebrae and associated vertebral numeration, *Eur. Spine J.* **27**, 995–1004 (2018), doi:10.1007/s00586-018-5554-8.
- ²G. P. Konin and D. M. Walz, Lumbosacral transitional vertebrae: classification, imaging findings, and clinical relevance, *AJNR Am. J. Neuroradiol.* **31**, 1778–1786 (2010), doi:10.3174/ajnr.A2036.

³Y. Deng et al., CTSpine1K: a large-scale dataset for spinal vertebrae segmentation in computed tomography, *Mach. Learn. Biomed. Imaging* **3**, 824–832 (2025), doi:10.59275/j.melba.2025-gf84.

⁴P. Liu et al., Deep learning to segment pelvic bones: large-scale CT datasets and baseline models, *Int. J. Comput. Assist. Radiol. Surg.* **16**, 749–756 (2021), doi:10.1007/s11548-021-02363-8.

⁵A. Sekuboyina et al., VerSe: a vertebrae labelling and segmentation benchmark for multi-detector CT images, *Med. Image Anal.* **73**, 102166 (2021), doi:10.1016/j.media.2021.102166.

⁶H. Liebl et al., A computed tomography vertebral segmentation dataset with anatomical variations and multi-vendor scanner data, *Sci. Data* **8**, 284 (2021), doi:10.1038/s41597-021-01060-0.

⁷L. Jin et al., RibSeg v2: a large-scale benchmark for rib labeling and anatomical centerline extraction, *IEEE Trans. Med. Imaging* **43**, 570–581 (2024), doi:10.1109/TMI.2023.3313627.

⁸K. Smith et al., Data from CT COLONOGRAPHY, The Cancer Imaging Archive, 2015, doi:10.7937/K9/TCIA.2015.NWTESAY1.

⁹K. Clark et al., The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository, *J. Digit. Imaging* **26**, 1045–1057 (2013), doi:10.1007/s10278-013-9622-7.

¹⁰H. Möller et al., Automated thoracolumbar stump rib detection and analysis in a large CT cohort, 2025, arXiv:2505.05004. doi:10.48550/arXiv.2505.05004.

¹¹J. Wasserthal et al., TotalSegmentator: robust segmentation of 104 anatomic structures in CT images, *Radiol. Artif. Intell.* **5**, e230024 (2023), doi:10.1148/ryai.230024.

¹²A. Schehr, J. Kim, and G. Schwing, OpenSpineConsortium: an open-source framework for medical student engagement in computational spine imaging research, *Cureus* (2026), Published 14 July 2026. doi:10.7759/cureus.112661.

¹³F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, *Nat. Methods* **18**, 203–211 (2021), doi:10.1038/s41592-020-01008-z.

¹⁴R. Vialle, N. Levassor, L. Rillardon, A. Templier, W. Skalli, and P. Guigui, Radiographic analysis of the sagittal alignment and balance of the spine in asymptomatic subjects, *J.*

471 Bone Joint Surg. Am. **87**, 260–267 (2005), doi:10.2106/JBJS.D.02043.

472 ¹⁵A. E. Castellvi, L. A. Goldstein, and D. P. K. Chan, Lumbosacral transitional vertebrae
 473 and their relationship with lumbar extradural defects, Spine (Phila. Pa. 1976) **9**, 493–495
 474 (1984), doi:10.1097/00007632-198407000-00014.

475 ¹⁶P. J. Pickhardt, B. D. Pooler, T. Lauder, A. M. del Rio, R. J. Bruce, and N. Binkley,
 476 Opportunistic screening for osteoporosis using abdominal computed tomography scans
 477 obtained for other indications, Ann. Intern. Med. **158**, 588–595 (2013), doi:10.7326/0003-
 478 4819-158-8-201304160-00003.

479 ¹⁷N. E. Epstein, A perspective on wrong level, wrong side, and wrong site spine surgery,
 480 Surg. Neurol. Int. **12**, 286 (2021), doi:10.25259/SNI.402_2021.

481 ¹⁸E. C. Benzel, *Biomechanics of Spine Stabilization*, Thieme, New York, 3rd edition, 2015.

482 ¹⁹E. Beaulieu et al., A review of finite element analysis in spine surgery decision-making, J.
 483 Clin. Med. **15**, 2584 (2026), doi:10.3390/jcm15072584.

Supplementary Material

CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

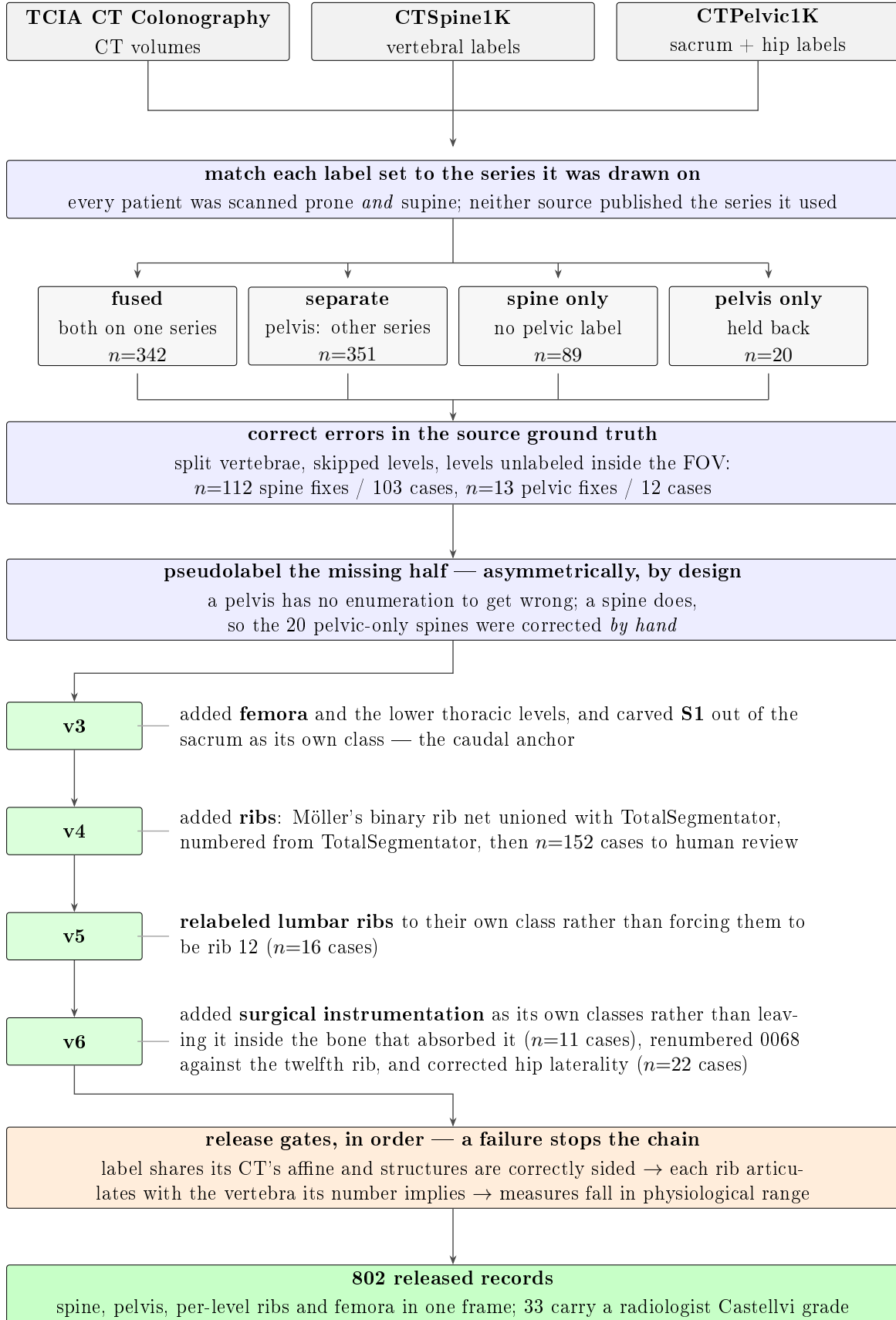


Figure S 1: How the dataset was built. Three public collections are combined patient by patient. Neither annotation source published the CT series its labels were drawn on, so each label set is matched to the series whose bone agrees with it. In 351 of those patients the two collections had drawn their outlines on *different scans of the same person*. Numbers marked *n* are scans, or corrections, at that step.